A counterexample to the printed sign conjecture in Montgomery–Smith–Semenov

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Source Statement

The source is S. J. Montgomery-Smith and E. M. Semenov, *Embeddings of rearrangement invariant spaces that are not strictly singular*, arXiv:math/0007058, Positivity 4 (2000), 397–404 [1]. On PDF page 2 it states Conjecture 3:

> Given $x_1, \dots, x_n \in L_1([0,1])$, and a rearrangement invariant space E on $[0,1]$, there exists signs $\epsilon_1, \dots, \epsilon_n = \pm 1$ such that
> $$\left\| \sum_{i=1}^{n} \epsilon_i x_i \right\|_{L_1} \geq c^{-1} \left\| \sum_{i=1}^{n} r_i \|x_i\|_1 \right\|_E .$$

It is clear that if E is a rearrangement invariant space on $[0,1]$ that contains G continuously, then the natural embedding $E \subset L_1$ is not strictly singular. Here, of course, the subspace on which the norms are equivalent is the span of the Rademacher functions.

For this reason, it is natural to pose the following conjecture.

CONJECTURE 2. *Let E be a rearrangement invariant space on $[0,1]$. Then the natural embedding $E \subset L_1$ is not strictly singular if and only if G embeds into E continuously.*

It will become apparent below that the following conjecture implies the previous one.

CONJECTURE 3. *Given $x_1, \dots, x_n \in L_1([0,1])$, and a rearrangement invariant space E on $[0,1]$, there exists signs $\epsilon_1, \dots, \epsilon_n = \pm 1$ such that*
$$\left\| \sum_{i=1}^{n} \epsilon_i x_i \right\|_{L_1} \geq c^{-1} \left\| \sum_{i=1}^{n} r_i \|x_i\|_1 \right\|_E .$$

Unfortunately we are not able to prove either of these conjectures without some additional hypotheses.

First, let us introduce some examples of rearrangement invariant spaces. A function $\Phi : \mathbb{R} \to [0, \infty)$ is called an *Orlicz function* if it is convex, even, and takes zero to zero. The *Orlicz space* L_Φ is the

Proof Intuition

The printed conjecture quantifies over all rearrangement invariant spaces. For $E = L_\infty$, the right side is the largest pointwise size of the Rademacher sum with positive coefficients. If the test functions themselves are the Rademachers, changing signs only changes which Rademacher sum appears on the left. Its L_1-size is of order $\sqrt{n}$, while its L_∞ size is exactly n.

Counterexample

Theorem 1. *Conjecture 3 of [1] is false as printed.*

Proof. Let $E = L_\infty[0,1]$. For $n \geq 1$, set $x_i = r_i$, where $(r_i)_{i=1}^n$ are the first n Rademacher functions on $[0,1]$. Then $\|x_i\|_1 = 1$ for all i, and hence

$$\left\| \sum_{i=1}^n r_i \|x_i\|_1 \right\|_{L_\infty} = \left\| \sum_{i=1}^n r_i \right\|_{L_\infty} = n,$$

because the finite Rademacher vector attains the all-+1 sign pattern on a dyadic atom of positive measure.

On the other hand, for any fixed signs $\epsilon_i = \pm 1$,

$$\left\| \sum_{i=1}^n \epsilon_i r_i \right\|_{L_1} \leq \left\| \sum_{i=1}^n \epsilon_i r_i \right\|_{L_2} = \sqrt{n}.$$

Thus the conjectured inequality would require

$$\sqrt{n} \geq c^{-1} n$$

for every n. Equivalently, $c \geq \sqrt{n}$ for every n, impossible for a fixed finite constant c. Therefore the printed Conjecture 3 fails. $\square$

Relation to Conjecture 2

This counterexample does not refute the paper's main Conjecture 2. A later paper of Hernandez, Ruiz, and Sanchiz records the corresponding classification for symmetric, equivalently rearrangement invariant, function spaces as known: the inclusion $E(\mu) \hookrightarrow L_1(\mu)$ is strictly singular if and only if $L_0^{\exp x^2}$ is not included in $E(\mu)$, citing Hernandez–Novikov–Semenov and Astashkin–Hernandez–Semenov [2].

Verification and Literature Check

The proof uses only Rademacher independence and Cauchy–Schwarz on the probability interval. Local run indexes were searched for arXiv:0007058, rearrangement invariant strict singularity, Rademacher spaces, and the exact title; no existing packet for this sign-conjecture counterexample was found. Web searches for the exact source title and strict-singular/Rademacher keywords surfaced the source arXiv page and related Rademacher-space literature, but no prior record of this printed Conjecture 3 counterexample.

Human Review Recommendation

Review as a likely valid literal counterexample to Conjecture 3. The main interpretive question is whether the authors intended an additional restriction on E, or intended a different norm on the left side; either restriction would be a different statement from the one printed.

References

[1] S. J. Montgomery-Smith and E. M. Semenov, *Embeddings of rearrangement invariant spaces that are not strictly singular*, Positivity 4 (2000), 397–404. https://arxiv.org/abs/math/0007058

[2] F. L. Hernandez, C. Ruiz, and M. Sanchiz, *Disjointly strictly singular inclusions between variable Lebesgue spaces*, arXiv:2406.14175, 2024. https://arxiv.org/abs/2406.14175